

Final Project Progress Report

Project scope update

The project scope has updated beyond the original proposal. Multiple ML models will now be developed and compared to identify the best predictor of colon cancer mortality. My project will now include the following machine learning models such as logistic regression, random forest, and gradient boosting. As well as model evaluation and other visualization models to showcase the model being extracted. This decision was made based on feedback received during my project proposal submission and reflects best practices in model evaluation by comparing models against each other producing a more reliable and strong results rather than relying on a single approach.

Data preprocessing is currently underway. Given the large file sizes involved, an initial data cleaning assessment is being conducted to identify missing values, outliers, and inconsistencies. As part of this process, several columns originally stored as object (dtype) have been converted to category (dtype) in a Jupyter Notebook environment. This conversion reduces memory usage significantly on large datasets and improves the efficiency of ML pipelines, particularly for tree-based models such as Random Forest and Gradient Boosting, which handles categorical features well.

Data sources

Since encountering difficulties with the original data extraction process, I have pivoted to a new API to better support the project's goal of predicting colon cancer mortality and will be updated in my data sources table. This updated API focuses on incorporating clinical trial interventions and their specific outcomes, providing a stronger evidence-based foundation for the machine learning models. By integrating these trial results, I can analyze how different treatments impact mortality rates across a wider range of predictive models. To manage this more complex dataset effectively, I am continuing to clean and optimize the data within Jupyter Notebooks, ensuring the large files are formatted for efficient processing and will be copied to a python file once I am done with preprocessing. I also found a new data source CDC WONDER API, which has proven to be more reliable, better documented, and more directly aligned with my project's need for mortality statistics.

Issues / difficulties

The primary challenge encountered during this phase is extracting data from the SEER API. Despite successfully obtaining an API key and constructing properly authenticated requests, multiple endpoints returned inconclusive data when populated and showed no relevance to my project end goals. I am asking for your approval or feedback on this situation.